



Bhabha University Bhopal (M.P.)

Scheme of Examination

Programme – Master of Technology

Branch – Civil Engineering

Specialization – Construction Technology & Management

First Semester

Session 2018-19

Sr .No.	Subject Code	Subject Name	Periods per week			Credits	Maximum Marks (Theory Slot)			Marks for End Sem. Examination		Total Marks
			L	T	P		End Sem. Exam.	Tests (Two)	Assignments /Quiz	End Sem. Practical /Viva	Practical Record/Assignment/Quiz /Presentation	
1	MA-102	Advanced Mathematics	3	1	-	4	70	20	10	-	-	100
2	CE-101	Construction Material	3	1	-	4	70	20	10	-	-	100
3	CE -102	Ad. Geotechnical Engg.	3	1	-	4	70	20	10	-	-	100
4	CE -103	Construction Technology	3	1	-	4	70	20	10	-	-	100
5	CE -104	Low cost Build materials & cont. Technology	3	1	-	4	70	20	10	-	-	100
6	CE -105	Lab-I Computer Workshop	-	-	6	6	-	-	-	90	60	150
7	CE -106	Lab-II Ad. Construction -I	-	-	6	6	-	-	-	90	60	150
		Total	15	5	12	32	350	100	50	180	120	800

L: Lecture T: Tutorial P: Practical

End semester exam will be of 3 hours duration



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M.TECH (CTM)

ADVANCE MATHEMATICS

Subject Code-MA102

Unit-1

Introduction of operation-research Mathematical Programming Techniques Linear programming (formulation, Graphical solution). Simplex method, BIG-M method. Duality theory in linear programming, Transportation & Assignment Problem.

Unit-2

Network Methods CPM, PERT, Dynamic Programming, Bellmon's Principle of optimality, Non linear Programming. Fibonacci method.

Unit-3

Game theory, Queuing system Genetic algorithm, ANN methods, Evolutionary algorithms.

Unit-4

Probability concepts , Additive and Multiplicative laws of Probability , Boye's theorem, statistical frequency distribution, Binomial, Poisson, Normal Distribution, confidence intervals, Tests of Significance for small & large samples. Testing of Hypothesis, Linear and Non linear regression.

Unit-5

Reliability Engineering Basic concepts of Reliability, Design for Reliability constraints and consideration Reliability and Mathematics, Components Reliability and Hazard models, System Reliability models.

Books & References Recommended:

1. Operation Research by Phillips & Ravindran
2. Operation Research by TAHA
3. Probability, Statistics & Decision in Civil Engineering by Benjamin & Cornell
4. optimization by S.S. Rao



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M.TECH (CTM)

Construction Materials

Subject Code -CE101

Unit-1

Material Science

Classification, Standardisation, Codification and Variety. Details of Micro Structure of Different construction Materials, Different effects on materials of construction.

Unit-2

Properties of Materials

Environmental Influences : Thermal effects : Effect of Chemicals, Fire resistance, Corrosion and Oxidation, Radiation. Properties of fresh & hardened concrete. Shrinkage & creep of concrete.

Unit-3

Concrete

Design and production of concrete ingredients, Additives and admixtures. Special concretes e.g. light weight, Heavy weight, Ready mix concrete, Fibre Reinforced concrete etc.

Unit-4

New Construction Materials

Polymer materials, Thermo - Plastic, Polymer Concrete, Composite materials, Ferrocement, Ferroconcrete, Building materials from Agricultural & Industrial wastes.

Unit-5

Quality control in construction

Various aspects, Principle of statistical quality control. Different techniques of materials and process Quality control, Destructive and non destructive Testing of Materials, I.S. and international procedures of testing.

References:

1. Ammer, D.A. Material Management Irwin Publishers Illionis, 1972.
2. White A.H. Engineering materials, MC Graw - Hill.
3. Deb. A., Engineerig materials, world press.
4. Billmeyer Jr. F.W. Text Book of Polymer Science, Interscience Publishers Inc.
5. Golding Brage Polymers and Resins Nortrand.
6. Schmidt A.X. & Marties CA "Principle of High Polymer Theory & Practice" MC Graw – Hill.
7. Stille, J.K. "Introduction to Polymer Chemistry" Johwiley.
8. Winding C.C. & Hiatt G.D. "Polymetric".



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M.TECH (CTM)

Advanced Geotechnical Engineering

Subject Code-CE102

Unit-1

Site Investigations & Stress Distribution in soils

Brief review of various methods of subsurface explorations, soil sampling, subsurface soundings, Geophysical explorations. Stress distribution beneath loaded areas by Boussinesq Westergaard's and Steinbrenner methods. Newmark's influence chart. Contact pressure distribution. Settlement analysis.

Unit-2

Well Foundations & Cofferdams

Types of caissons, Wells, and their design criteria. IS and IRC codes and their provisions. Tilt and Shift in wells and their rectifications. Types, Design data for cellular dams, stability analysis. interlock Stresses, Methods of design of cellular coffer dams.

Unit-3

Machine Foundations

Theory of Vibrations. Single and double degree of freedom system. Damped and undamped vibrations. Types of machine foundations, mass spring model of analysis. Apparent mass of soil. Design of block foundations for impact type of machinery. Indian standard on Design and Construction of Foundations for Reciprocating machines.

Unit-4

Foundations on Expansive Soils

Characteristics and treatment of expansive soils. Construction techniques in expansive soils. Use of under-reamed piles and their design criteria, CNS Layer techniques. Construction on collapsible soil

Unit-5

Rock Mechanics

Problems in Rock mechanics, Classification of rocks, physical, geological and Mechanical properties of rocks, mechanics of rock, deformation and fracture under load. The range and scope of Rock mechanics in relation to civil engineering projects.



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References

1. Soil Mechanics and Foundations –by Dr. B.C. Punamia, Er. Ashok K. Jain
2. Geotechnical Engineering –by Saurabh Kumar Soni
3. Advance Geotechnical Engineering : Soil-Structure –by C.S Desai and Musharraf Zaman
4. Advance Soil Mechanics –by Braja M. Das
5. Soil Mechanics in Engineering Practice –by Karl Terzaghi, Ralph B. Peck, Gholamreza Mesri
6. Foundation Engineering –by Ralph Brazelton
7. Fundamental of Soil Behaviour –by James Mitchell



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Construction Technology

Subject Code-CE103

Unit-1

Advanced Pavement Construction Techniques

Pavement Construction using Bitumen, Hot mix plant, Concrete Road Construction, Fibre Reinforced Pavement Construction, Low Cost Road Construction Techniques.

Unit-2

Form Work and Temporary Structures

Design and construction features of different types of Temporary Structures. Stationary and slip form work Techniques, Special features of insitu construction. Stripping and Removal of form works, Form works for special structures e.g. shells, bridges, towers etc.

Unit-3

Steel Construction

Shop and insitu construction techniques, different connections. High strength bolts, Clearances and Tolerances, Erection of steel structures like Bridges, Trusses Chimneys, Power Houses.

Unit-4

Prestressing

Plants, Equipment for Prestressed Construction, Different Techniques of Prestressing. Prestressing of Bridge girders, water tanks and special structures.

Unit-5

Construction Techniques of Heavy and Special Structures

Dams, Bridges, large spanroofs, high rise Buildings, off shore Platforms, Pipelines, Tunnels and other under ground structures, Safety measures in Construction.

References -

1. Construction Technology –by Mr. Roy Chudley & Roger Greend
2. Introduction to Construction Building –by Stephen Emmitt
3. Construction Technology –by Roy Chudley
4. Introduction to building –by Derek Osbourn
5. Barry's Introduction to Construction of Buildings –by Stephen Emmitt
6. Principle of Construction –by Roger Greeno



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M.TECH (CTM)

Low Cost Building Materials and Construction Technology

Subject Code-CE104

Unit-1

Concepts of low cost materials

Soil, Fly ash, ferrocement, Lime, Fibers, Stone Dust, Boulders and oversize metal, Bitumen etc.

Unit-2

Low cost building material products

(a) Walls - Stabilized and sun dried, soil blocks & bricks, Hollow concrete blocks, stone masonry blocks, Ferro-cement partitions.

(b) Roofs - Precast R.C. Plank & Joists roof, Precast channel roof, Precast L-panel roof, Precast Funicular shells, Ferrocement shells, Filler Slab, Seasal Fibre roof, Improved country tiles, Thatch roof.

Unit-3

Low cost construction Techniques and Equipment

(a) Techniques :- Rat trap bond construction, Precast R.C. and Ferrocement technique, Mud Technology.

(b) Equipments :- Brick moulding machine, Stabilised soil block making machine and plants for the manufacturing of concrete blocks.

(c) Low Cost Roads :-

Unit-4

Low cost sanitation

(a) Waste water disposal system

(b) Low cost sanitation for rural and urban areas

(c) Ferrocement Drains

Unit-5

Cost analysis and comparison

(a) Low cost materials

(b) Low cost techniques



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Reference :-

1. Hand Book of Low Cost Housing –by A.K. Lal
2. Low Cost Building Construction –by Ralph Wolfe
3. Low Cost Housing –by Deepak Kumar
4. Building an Affordable House –by Fernando Pages Ruiz
5. Project Cost Construction Control –by Roy Pilcher

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Lab-I Computer Workshop LAB-I (CE 105

Computational Facility (Hardware/Software):

1. STAAD - PRO
2. AUTO - CAD
3. MATLAB

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ADVANCE CONSTRUCTION LAB-I (CE 106)

List of Experiment

1. Tests on Aggregates

2. Tests on Cement

3. Determination of compressive strength of concrete with different cement grades.

4. Determination of workability of concrete by slump test

5. Determination of workability by compacting factor apparatus.

6. Determination of workability by Vee Bee consistometer.

7. Nondestructive testing of concrete by Rebound hammer test

8. Nondestructive testing of concrete by ultrasonic Method.

9. Test for the effect of admixtures on the concrete compressive strength

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<i>10. Testing of micro-concrete</i>
<i>11. Design of concrete mix.</i>

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